

**PROJECT TITLE: DEVELOPMENT AND PRETESTING OF AN ONLINE SELF  
SCREENING TOOL FOR INDIVIDUAL REQUIRING FURTHER ASSESSMENT FOR  
CARDIAC DISEASE, DIABETES MELLITUS AND CERVICAL CANCER**

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**B. ABSTRACT**

The burden of non-communicable diseases, especially diabetes mellitus, cardiac disease, and cancer, is increasing rapidly in Africa. In Uganda, many individuals with NCDs remain undiagnosed owing to limited awareness and low screening amongst the populations at risk. We propose to develop an online assessment screening tool for individuals to self-screen their risk of cardiac disease, diabetes mellitus and cervical cancer. The proposed tool combining screening questions assessing individual characteristics, behavioral risks factors and early symptoms to determine whether further evaluations by health workers are recommended or not. Through this tool, we will leverage the increased access to the internet, computers and mobile gadgets to increase access to screening services for these NCDs. The tool, if scaled-up could increase uptake of NCD screening services thus contributing to reduced burden of disease through early identification and treatment of preclinical stages of NCDs as well as secondary prevention of mortality and long-term complications of these conditions.

**C. BACKGROUND**

The burden of non-communicable diseases (NCDs), is increasing rapidly in Africa and it is projected that NCDs may account for 46 % of deaths in Sub Saharan Africa by 2030 (1). Cardiac disease (hypertension and ischemic heart disease), diabetes mellitus (DM) and cancer are of particular concern. Morbidity and mortality due to these conditions has increased owing to longer life expectancy and increase the prevalence of their risk factors. For example, many individuals especially those living in urban areas are engaged in sedentary work and they consume calorie-dense low fibre diets that predispose them to these conditions (2, 3). Moreover, NCDs in developing countries affect people at a younger age compared with those in developed countries thus significantly affecting their productivity (4).

In Uganda, many individuals with NCDs remain undiagnosed owing to limited awareness and low screening amongst the population at risk. According to the Uganda nationwide NCD risk factor survey conducted in 2014, the prevalence of hypertension in Uganda as high as 26.4% with only 7.7% of hypertensive persons aware of their hypertension(5). In a cross-sectional

survey done amongst 497 randomly selected participants from Uganda, 16.1% of rural residents and 7.6% of peri-urban residents had diabetes yet over 60% of DM patients were not aware they had it (6). These statistics point to a high burden of undiagnosed NCDs even amongst urbanized populations that are expected to have more access to information.

Like other NCDs, cancer in Uganda is characterized by late presentation with most individuals reporting or being diagnosed late when the cancer is more difficult to treat. Cervical cancer is the commonest gynaecological malignancy among women in Uganda and the leading cause of cancer deaths. With incidence and mortality rates of 54.8 and 40.5 cases per 100,000 women respectively, Uganda remains one of the high burden countries across the world (7). And yet this cancer is preventable through vaccination of girls and timely routine screening of women. The uptake of screening services remains low. In a cross-sectional study done amongst 900 women in Eastern Uganda, only 4.8% and almost half did so upon request by a health worker implying that the rate of voluntary screening was quite low(8).

We, therefore, propose to develop an online assessment screening tool for individuals to self-screen their risk of cardiac disease, diabetes mellitus and cervical cancer. The proposed tool containing questions with a mixture of individual characteristics, behavioural risks factors and early symptoms will compile the responses thereby determining whether further evaluation by a trained health worker is recommended or not. This tool will leverage the increased access to the internet, computers, and mobile gadgets to increase access to screening services for these NCDs. The tool if utilized and scaled-up could increase uptake of NCD screening services thus contributing to deduced burden of disease in two ways: First, if pre-disease states are identified, individuals can be empowered to tackle modifiable risk factors such overweight or obesity to prevent NCDs from manifesting clinically. Secondary, for individuals diagnosed with NCDs, mortality and long-term complications could be prevented through effective treatment to control these conditions.

#### **D. METHODS**

**Goal:** To increase uptake of screening and treatment services for cardiac disease, diabetes mellitus and cervical cancer.

**Objectives:**

1. To develop an online self-screening tool for cardiac diseases, diabetes and cervical cancer
2. To assess the acceptability of the tool and health seeking behavior for individuals eligible for additional evaluation (those screened in)

**Approach:**

**Phase 1 (Year 1): Development of the online application**

1. Desk review to compile risk criteria/algorithm for further screening for each condition  
For each condition, review of existing literature related to risk factors and causality for each condition will be conducted to identify areas that can be assessed using simple easy to interpret screening questions.
2. Development of user requirements for a screening tool  
The team will review the questions or criteria identified for each condition and construct screening algorithms.

3. Development of the online application  
The application will be developed and tested prior to the study.

**Phase 2 (Year 2-3): Cross-sectional study to pretest the tool and assess its effect on subsequent health-seeking behaviour.**

1. Research protocol writing and regulatory approvals  
A research protocol will be written to study the acceptability of the screening tool and its effect on subsequent health seeking behavior.
2. Research implementation  
The study will be implemented as per approved protocol.
3. Analysis and Dissemination of results  
The results will be disseminated to locally to NCD department of Ministry of health and other stakeholders. In addition, the results will be published in a scientific journal.

**E. PROPOSED OUTPUTS/DELIVERABLES**

**Phase 1 outputs:**

1. Risk criteria or algorithm for agreed-upon conditions
2. User requirements for self-screening tool
3. Online application

**Phase 2 outputs:**

1. Research proposal approved by IRB and UNCST
2. A cross-sectional study on acceptability of the tool by intended completed
3. Study report disseminated

**F. REFERENCES**

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## G. ITEMISED BUDGET

|                                                                                                                                    | Key Activities                                                           | Unit Cost (Ugx) | Unit Description                           | Number of units | Total Cost (Ugx)   |
|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------|--------------------------------------------|-----------------|--------------------|
| <b>Phase 1 (Year 1): Development of the self-screening tool</b>                                                                    | 1.1 Desk review to compile risk criteria/algorithm for further screening | 5,000,000       | Consultancy fees per month                 | 3               | 15,000,000         |
|                                                                                                                                    | 1.2. Meetings to develop user requirements for screening tool            | 3,000,000       | Venue, meeting refreshments and allowances | 3               | 9,000,000          |
|                                                                                                                                    | 1.3. Development of the online application                               | 30,000,000      | Application Development fees               | 1               | 30,000,000         |
|                                                                                                                                    | Sub Total for Phase 1                                                    |                 |                                            |                 | 54,000,000         |
| <b>Phase 2 (Year 2-3): Cross-sectional study to pretest the tool and assess its effect on subsequent health seeking behaviour.</b> | 2. 1. Research protocol writing                                          | 5,000,000       | Consultancy fees per month                 | 3               | 15,000,000         |
|                                                                                                                                    | 2.2. Regulatory approvals for research protocol                          | 3,000,000       | IRB and UNCST fees                         | 1               | 3,000,000          |
|                                                                                                                                    | 2.3. Research implementation                                             |                 |                                            |                 |                    |
|                                                                                                                                    | 2.3.1. Stationery                                                        | 500,000         | monthly estimate                           | 24              | 12,000,000         |
|                                                                                                                                    | 2.3.2. Survey CTO                                                        | 400,000         | monthly subscription                       | 12              | 4,800,000          |
|                                                                                                                                    | 2.3.3. Survey gadgets                                                    | 1,000,000       | Tablets                                    | 20              | 20,000,000         |
|                                                                                                                                    | 2.3.4. Field research assistants                                         | 200,000         | Man days                                   | 120             | 24,000,000         |
|                                                                                                                                    | 2.3.5 Transportation to field                                            | 300,000         | Fueled, hired car rate                     | 100             | 30,000,000         |
|                                                                                                                                    | 2.3.6 Consultancy fees                                                   | 800,000         | Daily Consultancy fees                     | 40              | 32,000,000         |
|                                                                                                                                    | 3. Analysis and Dissemination of results                                 |                 |                                            |                 |                    |
|                                                                                                                                    | 3.1. Data cleaning and Analysis                                          | 5,000,000       | Monthly Consultancy fees                   | 3               | 15,000,000         |
|                                                                                                                                    | 3.2. Report writing                                                      | 5,000,000       | Consultancy fees per month                 | 2               | 10,000,000         |
|                                                                                                                                    | 3.3. Manuscript writing                                                  | 5,000,000       | Consultancy fees per month                 | 3               | 15,000,000         |
|                                                                                                                                    | 3.4. Publication fees                                                    | 8,000,000       | Scientific journal publication fees        | 1               | 8,000,000          |
|                                                                                                                                    | Sub Total for Phase 2                                                    |                 |                                            |                 | 188,800,000        |
|                                                                                                                                    | <b>TOTAL PROJECT BUDGET</b>                                              |                 |                                            |                 | <b>252,800,000</b> |

